

위스퍼 음성 인식 기초

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-
- openai 설치
 - whisper 설치
 - ffmpeg 설치

설치 및 시작하기

```
In [1]: #!pip install -U openai-whisper
```

```
In [ ]: # !winget install ffmpeg -e --accept-source-agreements --accept-package-agreements
```

'msstore' 원본을 사용하려면 다음 계약을 확인해야 합니다.

Terms of Transaction: <https://aka.ms/microsoft-store-terms-of-transaction>

원본이 제대로 작동하려면 현재 컴퓨터의 두 글자 지리적 지역을 백 엔드 서비스로 보내야 합니다(예: "미국").

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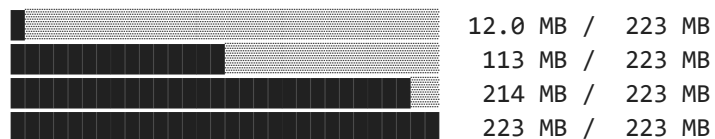
찾음 FFmpeg [Gyan.FFmpeg] 버전 8.0.1

이 응용 프로그램의 라이선스는 그 소유자가 사용자에게 부여했습니다.

Microsoft는 타사 패키지에 대한 책임을 지지 않고 라이선스를 부여하지도 않습니다.

다운로드 중 https://github.com/GyanD/codexffmpeg/releases/download/8.0.1/ffmpeg-8.0.1-full_build.zip

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설치 관리자 해시를 확인했습니다.

보관 파일을 추출하는 중...

보관 파일을 추출했습니다.

패키지 설치를 시작하는 중...

명령줄 별칭이 추가되었습니다. "ffmpeg"

명령줄 별칭이 추가되었습니다. "ffplay"

명령줄 별칭이 추가되었습니다. "ffprobe"

경로 환경 변수 수정됨; 셸을 다시 시작하여 새 값을 사용합니다.

설치 성공

```
In [1]: import whisper

print(whisper.__version__)
```

20250625

```
In [7]: # 오디오 파일 다운로드
!curl -o Winston_Church.ogg https://upload.wikimedia.org/wikipedia/commons/7/75/Winston_Churchill_-_Be_Ye_Men_of_Valour.ogg
```

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload Upload	Total	Spent	Left	Speed
0	0	0	0	0	--:--:--	--:--:--	0
0	0	0	0	0	--:--:--	--:--:--	0
100	2408k	100	2408k	0	0:00:01	0:00:01	2129k

base 모델 사용시

```
In [ ]: # 개인 폴더에서 작업하시는 경우 경로에 한글이 포함되면 에러가 날 확률이 높습니다.
# 1분 이상의 시간이 걸리는 경우가 많습니다.

!whisper Winston_Church.ogg --model base
```

Detecting language using up to the first 30 seconds. Use `--language` to specify the language

Detected language: English

[00:00.000 --> 00:07.680] I speak to you for the first time as Prime Minister in a solemn hour for the life of our country,
[00:08.320 --> 00:14.880] of our empire, of our allies, and above all of the cause of freedom.
[00:16.640 --> 00:20.240] A tremendous battle is raging in France and flanders.
[00:21.920 --> 00:27.920] The Germans by a remarkable combination of air bombing and heavily armored tanks
[00:28.720 --> 00:35.520] have broken through the French defenses north of the Maginot line and strong columns of
[00:35.520 --> 00:42.080] their armored vehicles are ravaging the open country which for the first day or two was without
[00:42.080 --> 00:48.640] defenders. They have penetrated deeply and spread alarm and confusion in their track.
[00:50.000 --> 00:56.480] Behind them there are now appearing infantry in lorries and behind them again the large masses
[00:56.480 --> 01:04.400] are moving forward. The regroupment of the French armies to make head again and also to strike
[01:04.400 --> 01:13.360] at this intruding wedge has been proceeding for several days largely assisted by the magnificent
[01:13.360 --> 01:21.520] efforts of the Royal Air Force. We must not allow ourselves to be intimidated by the presence
[01:21.520 --> 01:27.920] of the armored vehicles in unexpected places behind our lines. If they are behind our front
[01:28.640 --> 01:37.280] the French are also at many points fighting actively behind there. Both sides are therefore in an
[01:37.280 --> 01:45.520] extremely dangerous position and if the French army and our own army are well handled and I believe
[01:45.520 --> 01:52.160] they will be if the French retain that genius for recovery and counterattack which they have
[01:52.880 --> 02:01.120] so long been famous and if the British army shows the doggyed endurance and solid fighting power
[02:01.120 --> 02:08.560] of which there have been so many examples in the past then a sudden transformation of the scene
[02:08.640 --> 02:16.800] might spring into being. It would be foolish however to disguise the gravity of the hour.
[02:18.480 --> 02:24.160] It would be still more foolish to loot heart and courage or to suppose that well trained well
[02:24.160 --> 02:31.360] equipped armies numbering three or four millions of men can be overcome in the space of a few weeks
[02:31.360 --> 02:37.600] or even months by a scoop or raid of mechanized vehicles however formidable.
[02:38.800 --> 02:44.480] We may look with confidence to the stabilization of the front in front and to the general
[02:44.480 --> 02:50.240] engagement of the masses which will enable the qualities of the French and British soldiers
[02:50.240 --> 02:54.400] to be matched squarely against those of their adversaries.
[02:54.880 --> 03:03.600] For myself I have invincible confidence in the French army and its leaders only a very small part
[03:03.600 --> 03:11.440] of that splendid army and yet been heavily engaged and only a very small part of France had yet
[03:11.440 --> 03:17.840] been invaded. There is good evidence to show that practically the whole of the specialized and
[03:17.840 --> 03:23.840] mechanized forces of the enemy have been already thrown into the battle and we know that very
[03:23.920 --> 03:32.000] heavy losses have been inflicted upon them. No officer or man, no brigade or division which
[03:32.000 --> 03:37.600] grapples at close quarters with the enemy wherever encountered can fail to make a worthy
[03:37.600 --> 03:44.800] contribution to the general result. The armies must cast away the idea of resisting attack
[03:44.800 --> 03:52.080] behind concrete lines or natural obstacles and must realize that mastery can only be regained
[03:52.160 --> 03:59.440] by furious and unrelenting assault and this spirit must not only animate the high command
[03:59.440 --> 04:09.120] but must inspire every fighting man. In the air, often at serious odds, often at odds,
[04:09.120 --> 04:15.760] to the two who thought overwhelming, we have been clawing down three or four to one of our enemy
[04:17.040 --> 04:21.680] and the relative balance of the British and German air forces is now considerably more
[04:21.680 --> 04:28.480] favorable to us than at the beginning of the battle. In cutting down the German bombers,
[04:28.480 --> 04:34.720] we are fighting our own battle as well as that of France. My confidence in our ability to fight
[04:34.720 --> 04:40.400] it out to the finish with the German air force has been strengthened by the fierce encounters

[04:40.400 --> 04:47.200] which have taken place and are taking place. At the same time, our heavy bombers are striking
[04:47.280 --> 04:54.240] nightly at the taproot of German mechanized power and have already inflicted serious damage
[04:54.240 --> 05:00.720] upon the oil refineries on which the Nazi effort to dominate the world directly depends.
[05:02.480 --> 05:07.600] We must expect that as soon as stability is reached on the western front,
[05:08.400 --> 05:14.320] the bulk of that hideous apparatus of aggression which dashed Holland into ruin and slavery
[05:14.400 --> 05:22.880] in a few days will be turned upon us. I am sure I speak for all when I say we are ready to face it,
[05:23.840 --> 05:31.600] to endure it and to retaliate against it, to any extent that the unwritten laws of war permit.
[05:32.880 --> 05:39.600] There will be many men and many women in this island who when the ordeal comes upon them
[05:39.680 --> 05:46.480] and come it will will feel comfort and even a pride that they are sharing the perils of our
[05:46.480 --> 05:53.200] leds at the front. Soldiers, sailors and airmen, God bless them and are drawing away for them a
[05:53.200 --> 06:01.280] part at least of the onslaught they have to bear. He is not this the appointed time for all to make
[06:01.280 --> 06:08.320] the utmost exertions in their power. If the battle is to be won, we must provide our men with
[06:08.320 --> 06:15.040] ever increasing quantities of the weapons and ammunition they need. We must have and have quickly
[06:15.040 --> 06:21.920] more aeroplanes, more tanks, more shells, more guns. There is imperious need for these vital
[06:21.920 --> 06:28.400] munitions. They increase our strength against the powerfully armed enemy. They replace the wastage
[06:28.400 --> 06:36.000] of the obstinate struggle and the knowledge that wastage will speedily be replaced enables us to
[06:36.000 --> 06:43.280] draw more readily upon our reserves and throw them in now that everything counts so much. Our task
[06:43.280 --> 06:50.080] is not only to win the battle but to win the war. After this battle in France abates its force,
[06:51.040 --> 06:58.640] there will come the battle for our islands, for all that Britain is and all that Britain means
[06:59.360 --> 07:06.560] that will be the struggle. In that supreme emergency, we shall not hesitate to take every step,
[07:06.560 --> 07:14.080] even the most drastic to call forth from our people the last ounce and the last inch of effort
[07:14.080 --> 07:20.240] of which they are capable. The interests of property, the hours of labor are nothing compared to
[07:20.240 --> 07:25.200] the struggle for life and honor, for right and freedom to which we have vowed ourselves.
[07:25.600 --> 07:32.720] I have received from the chiefs of the French Republic and in particular from its
[07:33.680 --> 07:41.760] indomitable Prime Minister, Monsieur Reino, the most sacred pledges that whatever happens they will
[07:41.760 --> 07:50.800] fight to the end be it bitter nor be it glorious. No, if we fight to the end it can only be gloriou
s.
[07:51.760 --> 07:57.920] Having received its majesty's commission, I have formed an administration of men and women
[07:57.920 --> 08:04.640] of every party and of almost every point of view. We have differed and quarreled in the past
[08:05.600 --> 08:14.080] that now one bound unite us at all. The wage war until victory is won and never to surrender
[08:14.160 --> 08:19.840] ourselves to servitude and shame, whatever the cost and the agony may be.
[08:21.120 --> 08:27.520] This is one of the most awestriving periods in the long history of France and Britain.
[08:28.160 --> 08:36.000] It is also beyond doubt the most sublime, side by side, unaided except by their
[08:36.000 --> 08:41.680] kitten kin in the great dominions and by the wide empires which rest beneath their shield,
[08:42.000 --> 08:49.840] side by side the British and French peoples have advanced to rescue not only Europe but mankind
[08:50.480 --> 08:58.080] from the foulest and most soul-destroying tyranny which has ever darkened and stained the pages
[08:58.080 --> 09:09.440] of history. Behind us, behind the armies and fleets of Britain and France gather a group of
[09:09.440 --> 09:17.120] shattered states and bludgeoned races, the checks, the polls, the Norwegians, the Danes,
[09:17.120 --> 09:24.960] the Dutch, the Belgians, upon all of whom the long night of barbarism will descend unbroken
[09:24.960 --> 09:32.800] even by a star of hope unless we conquer as conquer we must, as conquer we shall.
[09:32.960 --> 09:36.880] Today is Trinity Sunday.

```
[09:38.400 --> 09:46.400] Century to go words were written to be a call and a spur to the faithful servants of truth and justice.
```

```
[09:48.640 --> 09:56.560] Arm yourselves and be men of valor and be in readiness for the conflict for it is better for us
```

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[09:56.560 --> 10:04.880] to perish in battle than to look upon the outrage of our nation and our altars as the will of God
```

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[10:04.880 --> 10:09.200] is in heaven, even so, let him do.
```

```
c:\ProgramData\anaconda3\Lib\site-packages\whisper\transcribe.py:132: UserWarning: FP16 is not supported on CPU; using FP32 instead
```

```
warnings.warn("FP16 is not supported on CPU; using FP32 instead")
```

medium 모델 사용 후, 음성 스크립트 만들기

```
In [ ]: # medium은 10분 이상, Large는 그 이상 걸릴 수 있으므로 더 정확하게 음성을 뽑아내고 싶다는 경우만 테스트 해 봅시다.  
# medium 모델 사용  
# !whisper Winston_Church.ogg --model medium  
  
# Large 모델 사용  
# !whisper src/whisper/audio/Winston_Church.ogg --model Large
```

Detecting language using up to the first 30 seconds. Use `--language` to specify the language

Detected language: English

[00:00.000 --> 00:08.400] I speak to you for the first time as Prime Minister in a solemn hour for the life of our country,
[00:08.400 --> 00:16.640] of our empire, of our allies, and above all, of the cause of freedom.
[00:16.640 --> 00:21.920] A tremendous battle is raging in France and Flanders.
[00:21.920 --> 00:28.720] The Germans, by a remarkable combination of air bombing and heavily armoured tanks,
[00:28.720 --> 00:35.840] have broken through the French defences north of the Maginot Line, and strong columns of their
[00:35.840 --> 00:42.080] armoured vehicles are ravaging the open country, which for the first day or two was without
[00:42.080 --> 00:48.640] defenders. They have penetrated deeply and spread alarm and confusion in their tracks.
[00:50.000 --> 00:56.480] Behind them there are now appearing infantry in lorries, and behind them again the large masses
[00:56.480 --> 01:04.960] are moving forward. The regroupment of the French armies to make head against and also to strike at
[01:04.960 --> 01:13.360] this intruding wedge has been proceeding for several days, largely assisted by the magnificent
[01:13.360 --> 01:21.600] efforts of the Royal Air Force. We must not allow ourselves to be intimidated by the presence of
[01:21.600 --> 01:27.920] these armoured vehicles in unexpected places behind our lines. If they are behind our front,
[01:28.560 --> 01:36.960] the French are also at many points fighting actively behind theirs. Both sides are therefore
[01:36.960 --> 01:44.240] in an extremely dangerous position, and if the French army and our own army are well handled,
[01:44.800 --> 01:51.280] as I believe they will be, if the French retain that genius for recovery and counter-attack
[01:51.280 --> 01:58.480] for which they have so long been famous, and if the British army shows the dogged endurance
[01:59.040 --> 02:05.040] and solid fighting power of which there have been so many examples in the past, then
[02:06.000 --> 02:14.560] a sudden transformation of the scene might spring into being. It would be foolish, however,
[02:14.560 --> 02:21.360] to disguise the gravity of the hour. It would be still more foolish to lose heart and courage,
[02:21.920 --> 02:27.520] or to suppose that well-trained, well-equipped armies numbering three or four millions of men
[02:28.240 --> 02:35.920] can be overcome in the space of a few weeks or even months by a scoop or raid of mechanised
[02:35.920 --> 02:42.000] vehicles, however, formidable. We may look with confidence to the stabilisation of the front in
[02:42.000 --> 02:48.000] France, and to the general engagement of the masses, which will enable the qualities of the
[02:48.000 --> 02:54.400] French and British soldiers to be matched squarely against those of their adversaries.
[02:55.840 --> 03:03.360] For myself, I have invincible confidence in the French army and its leaders. Only a very small
[03:03.360 --> 03:11.040] part of that splendid army has yet been heavily engaged, and only a very small part of France
[03:11.040 --> 03:17.600] has yet been invaded. There is good evidence to show that practically the whole of the specialised
[03:17.600 --> 03:23.600] and mechanised forces of the enemy have been already thrown into the battle, and we know that
[03:23.600 --> 03:31.120] very heavy losses have been inflicted upon them. No officer or man, no brigade or division,
[03:31.680 --> 03:37.600] which grapples at close quarters with the enemy wherever encountered, can fail to make a worthy
[03:37.600 --> 03:44.800] contribution to the general result. The armies must cast away the idea of resisting attack
[03:44.800 --> 03:52.080] behind concrete lines or natural obstacles, and must realise that mastery can only be regained
[03:52.080 --> 03:59.360] by furious and unrelenting assault, and this spirit must not only animate the high command,
[03:59.440 --> 04:09.040] but must inspire every fighting man. In the air, often at serious odds, often at odds,
[04:09.040 --> 04:15.760] to the two who thought overwhelming, we have been clawing down three or four to one of our enemy,
[04:17.040 --> 04:21.680] and the relative balance of the British and German air forces is now considerably more
[04:21.680 --> 04:28.480] favourable to us than at the beginning of the battle. In cutting down the German bombers,
[04:28.480 --> 04:34.320] we are fighting our own battle, as well as that of France. My confidence in our ability
[04:34.320 --> 04:39.520] to fight it out to the finish with the German air force has been strengthened by the fierce

[04:39.520 --> 04:46.800] encounters which have taken place and are taking place. At the same time, our heavy bombers are
[04:46.800 --> 04:52.960] striking nightly at the taproot of German mechanised power, and have already inflicted
[04:52.960 --> 05:00.240] serious damage upon the oil refineries on which the Nazi effort to dominate the world directly
[05:00.240 --> 05:09.200] depend. We must expect that as soon as stability is reached on the Western front, the bulk of that
[05:09.200 --> 05:15.520] hideous apparatus of aggression, which dashed Holland into ruin and slavery in a few days,
[05:15.600 --> 05:22.880] will be turned upon us. I am sure I speak for all when I say we are ready to face it,
[05:23.840 --> 05:31.600] to endure it, and to retaliate against it to any extent that the unwritten laws of war permit.
[05:32.880 --> 05:39.680] There will be many men and many women in this island who, when the ordeal comes upon them,
[05:39.680 --> 05:46.480] thence come it will, will feel comfort and even a pride that they are sharing the perils of our
[05:46.480 --> 05:53.200] lads at the front, soldiers, sailors and airmen, God bless them, and are drawing away for them a
[05:53.200 --> 06:01.280] part at least of the onslaught they have to bear. Is not this the appointed time for all to make
[06:01.280 --> 06:08.640] the utmost exertions in their power? If the battle is to be won, we must provide our men with ever
[06:08.640 --> 06:15.040] increasing quantities of the weapons and ammunition they need. We must have, and have quickly,
[06:15.040 --> 06:22.480] more aeroplanes, more tanks, more shells, more guns. There is imperious need for these vital munitions.
[06:23.600 --> 06:28.640] They increase our strength against the powerfully armed enemy, they replace the wastage of the
[06:28.640 --> 06:36.480] obstinate struggle, and the knowledge that wastage will speedily be replaced, enables us to draw
[06:36.480 --> 06:41.360] more readily upon our reserves and throw them in now that everything counts so much.
[06:42.560 --> 06:49.760] Our task is not only to win the battle, but to win the war. After this battle in France abates its
[06:49.760 --> 06:58.640] force, there will come the battle for our islands, for all that Britain is and all that Britain means.
[06:59.360 --> 07:05.760] That will be the struggle. In that supreme emergency, we shall not hesitate to take every
[07:05.760 --> 07:13.520] step, even the most drastic, to call forth from our people the last ounce and the last inch of
[07:13.520 --> 07:19.600] effort of which they are capable. The interests of property, the hours of labour, are nothing
[07:19.600 --> 07:25.120] compared to the struggle for life and honour, for right and freedom to which we have bowed ourselves.
[07:26.560 --> 07:32.720] I have received from the Chiefs of the French Republic, and in particular from its
[07:33.680 --> 07:41.280] indomitable Prime Minister, Monsieur Reynaud, the most sacred pledges that whatever happens,
[07:41.280 --> 07:49.040] they will fight to the end, be it bitter or be it glorious. Nay, if we fight to the end,
[07:49.040 --> 07:56.320] it can only be glorious. Having received His Majesty's commission, I have formed an
[07:56.320 --> 08:01.040] administration of men and women of every party and of almost every point of view.
[08:02.000 --> 08:07.840] We have differed and quarrelled in the past that now one bond unites at all,
[08:08.800 --> 08:16.400] to wage war until victory is won, and never to surrender ourselves to servitude and shame,
[08:17.120 --> 08:24.640] whatever the cost and the agony may be. This is one of the most awe-striking periods
[08:25.280 --> 08:31.520] in the long history of France and Britain. It is also, beyond doubt, the most sublime.
[08:32.480 --> 08:38.160] Side by side, unaided except by their kith and kin in the great dominions,
[08:38.160 --> 08:44.640] and by the wide empires which rest beneath their shield, side by side the British and
[08:44.640 --> 08:51.840] French peoples have advanced to rescue not only Europe, but mankind from the foulest
[08:51.840 --> 08:58.720] and most soul-destroying tyranny which has ever darkened and stained the pages of history.
[08:59.920 --> 09:07.120] Behind them, behind us, behind the armies and fleets of Britain and France,
[09:07.920 --> 09:16.000] gather a group of shattered states and bludgeoned races, the Czechs, the Poles, the Norwegians,

[09:16.000 --> 09:23.200] the Danes, the Dutch, the Belgians, upon all of whom the long night of barbarism will descend,
[09:23.760 --> 09:32.800] unbroken even by a star of hope, unless we conquer, as conquer we must, as conquer we shall.
[09:34.960 --> 09:42.960] Today is Trinity Sunday. Centuries ago words were written to be a call and a spur
[09:42.960 --> 09:46.400] to the faithful servants of truth and justice.
[09:48.640 --> 09:54.560] Arm yourselves and be ye men of valor, and be in readiness for the conflict,
[09:55.280 --> 10:00.800] for it is better for us to perish in battle than to look upon the outrage of our nation
[10:00.800 --> 10:07.920] and our altars, as the will of God is in heaven. Even so, let him do.

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1%		10.4M/1.42G [00:00<00:13, 109MiB/s]
1%		21.6M/1.42G [00:00<00:13, 114MiB/s]
2%		32.9M/1.42G [00:00<00:12, 116MiB/s]
3%		44.0M/1.42G [00:00<00:12, 116MiB/s]
4%		55.0M/1.42G [00:00<00:12, 116MiB/s]
5%		66.1M/1.42G [00:00<00:12, 115MiB/s]
5%		77.0M/1.42G [00:00<00:12, 115MiB/s]
6%		88.0M/1.42G [00:00<00:12, 115MiB/s]
7%		99.1M/1.42G [00:00<00:12, 115MiB/s]
8%		110M/1.42G [00:01<00:12, 116MiB/s]
8%		122M/1.42G [00:01<00:12, 117MiB/s]
9%		133M/1.42G [00:01<00:11, 117MiB/s]
10%		144M/1.42G [00:01<00:11, 117MiB/s]
11%		155M/1.42G [00:01<00:11, 117MiB/s]
11%		167M/1.42G [00:01<00:11, 118MiB/s]
12%		178M/1.42G [00:01<00:11, 118MiB/s]
13%		189M/1.42G [00:01<00:11, 117MiB/s]
14%		200M/1.42G [00:01<00:11, 117MiB/s]
15%		211M/1.42G [00:01<00:11, 115MiB/s]
15%		223M/1.42G [00:02<00:11, 116MiB/s]
16%		234M/1.42G [00:02<00:11, 116MiB/s]
17%		245M/1.42G [00:02<00:10, 117MiB/s]
18%		256M/1.42G [00:02<00:10, 117MiB/s]
18%		268M/1.42G [00:02<00:10, 117MiB/s]
19%		279M/1.42G [00:02<00:10, 118MiB/s]
20%		290M/1.42G [00:02<00:10, 117MiB/s]
21%		301M/1.42G [00:02<00:10, 118MiB/s]
21%		313M/1.42G [00:02<00:10, 118MiB/s]
22%		324M/1.42G [00:02<00:10, 118MiB/s]
23%		335M/1.42G [00:03<00:09, 118MiB/s]
24%		346M/1.42G [00:03<00:09, 118MiB/s]
25%		358M/1.42G [00:03<00:09, 117MiB/s]
25%		369M/1.42G [00:03<00:09, 118MiB/s]
26%		380M/1.42G [00:03<00:09, 118MiB/s]
27%		391M/1.42G [00:03<00:09, 118MiB/s]
28%		403M/1.42G [00:03<00:09, 117MiB/s]
28%		414M/1.42G [00:03<00:09, 117MiB/s]
29%		425M/1.42G [00:03<00:09, 116MiB/s]
30%		436M/1.42G [00:03<00:09, 116MiB/s]

31%	██████████	447M/1.42G [00:04<00:09, 114MiB/s]
31%	██████████	458M/1.42G [00:04<00:09, 115MiB/s]
32%	██████████	469M/1.42G [00:04<00:08, 116MiB/s]
33%	██████████	481M/1.42G [00:04<00:08, 116MiB/s]
34%	██████████	492M/1.42G [00:04<00:08, 116MiB/s]
35%	██████████	503M/1.42G [00:04<00:08, 117MiB/s]
35%	██████████	514M/1.42G [00:04<00:08, 117MiB/s]
36%	██████████	525M/1.42G [00:04<00:08, 117MiB/s]
37%	██████████	537M/1.42G [00:04<00:08, 117MiB/s]
38%	██████████	548M/1.42G [00:04<00:08, 117MiB/s]
38%	██████████	559M/1.42G [00:05<00:08, 117MiB/s]
39%	██████████	570M/1.42G [00:05<00:07, 118MiB/s]
40%	██████████	582M/1.42G [00:05<00:07, 118MiB/s]
41%	██████████	593M/1.42G [00:05<00:07, 118MiB/s]
41%	██████████	604M/1.42G [00:05<00:07, 116MiB/s]
42%	██████████	615M/1.42G [00:05<00:07, 116MiB/s]
43%	██████████	626M/1.42G [00:05<00:07, 116MiB/s]
44%	██████████	638M/1.42G [00:05<00:07, 116MiB/s]
45%	██████████	649M/1.42G [00:05<00:07, 117MiB/s]
45%	██████████	660M/1.42G [00:05<00:07, 117MiB/s]
46%	██████████	671M/1.42G [00:06<00:07, 117MiB/s]
47%	██████████	682M/1.42G [00:06<00:06, 117MiB/s]
48%	██████████	694M/1.42G [00:06<00:06, 117MiB/s]
48%	██████████	705M/1.42G [00:06<00:06, 117MiB/s]
49%	██████████	716M/1.42G [00:06<00:06, 117MiB/s]
50%	██████████	727M/1.42G [00:06<00:06, 117MiB/s]
51%	██████████	739M/1.42G [00:06<00:07, 103MiB/s]
51%	██████████	750M/1.42G [00:06<00:06, 107MiB/s]
52%	██████████	761M/1.42G [00:06<00:06, 109MiB/s]
53%	██████████	772M/1.42G [00:06<00:06, 111MiB/s]
54%	██████████	783M/1.42G [00:07<00:06, 113MiB/s]
54%	██████████	794M/1.42G [00:07<00:06, 114MiB/s]
55%	██████████	805M/1.42G [00:07<00:05, 115MiB/s]
56%	██████████	817M/1.42G [00:07<00:05, 116MiB/s]
57%	██████████	828M/1.42G [00:07<00:05, 116MiB/s]
58%	██████████	839M/1.42G [00:07<00:05, 117MiB/s]
58%	██████████	850M/1.42G [00:07<00:05, 117MiB/s]
59%	██████████	862M/1.42G [00:07<00:05, 117MiB/s]
60%	██████████	873M/1.42G [00:07<00:05, 117MiB/s]
61%	██████████	884M/1.42G [00:07<00:05, 117MiB/s]
61%	██████████	895M/1.42G [00:08<00:05, 117MiB/s]

62%		906M/1.42G [00:08<00:04, 117MiB/s]
63%		918M/1.42G [00:08<00:04, 118MiB/s]
64%		929M/1.42G [00:08<00:04, 118MiB/s]
65%		940M/1.42G [00:08<00:04, 118MiB/s]
65%		951M/1.42G [00:08<00:04, 118MiB/s]
66%		963M/1.42G [00:08<00:04, 118MiB/s]
67%		974M/1.42G [00:08<00:04, 117MiB/s]
68%		985M/1.42G [00:08<00:04, 118MiB/s]
68%		996M/1.42G [00:08<00:04, 117MiB/s]
69%		0.98G/1.42G [00:09<00:04, 112MiB/s]
70%		0.99G/1.42G [00:09<00:04, 113MiB/s]
71%		1.01G/1.42G [00:09<00:03, 115MiB/s]
71%		1.02G/1.42G [00:09<00:03, 116MiB/s]
72%		1.03G/1.42G [00:09<00:03, 116MiB/s]
73%		1.04G/1.42G [00:09<00:03, 117MiB/s]
74%		1.05G/1.42G [00:09<00:03, 117MiB/s]
75%		1.06G/1.42G [00:09<00:03, 117MiB/s]
75%		1.07G/1.42G [00:09<00:03, 117MiB/s]
76%		1.08G/1.42G [00:10<00:03, 117MiB/s]
77%		1.09G/1.42G [00:10<00:03, 118MiB/s]
78%		1.10G/1.42G [00:10<00:02, 118MiB/s]
78%		1.12G/1.42G [00:10<00:02, 118MiB/s]
79%		1.13G/1.42G [00:10<00:02, 118MiB/s]
80%		1.14G/1.42G [00:10<00:02, 118MiB/s]
81%		1.15G/1.42G [00:10<00:02, 118MiB/s]
81%		1.16G/1.42G [00:10<00:02, 118MiB/s]
82%		1.17G/1.42G [00:10<00:02, 116MiB/s]
83%		1.18G/1.42G [00:10<00:02, 112MiB/s]
84%		1.19G/1.42G [00:11<00:02, 114MiB/s]
85%		1.20G/1.42G [00:11<00:02, 113MiB/s]
85%		1.21G/1.42G [00:11<00:01, 114MiB/s]
86%		1.22G/1.42G [00:11<00:01, 115MiB/s]
87%		1.24G/1.42G [00:11<00:01, 116MiB/s]
88%		1.25G/1.42G [00:11<00:01, 116MiB/s]
88%		1.26G/1.42G [00:11<00:01, 116MiB/s]
89%		1.27G/1.42G [00:11<00:01, 117MiB/s]
90%		1.28G/1.42G [00:11<00:01, 117MiB/s]
91%		1.29G/1.42G [00:11<00:01, 117MiB/s]
91%		1.30G/1.42G [00:12<00:01, 117MiB/s]
92%		1.31G/1.42G [00:12<00:01, 117MiB/s]
93%		1.32G/1.42G [00:12<00:00, 117MiB/s]

```
c:\ProgramData\anaconda3\Lib\site-packages\whisper\transcribe.py:132: UserWarning: FP16 is not supported on CPU; using FP32 instead
  warnings.warn("FP16 is not supported on CPU; using FP32 instead")
```

```
In [4]: # !whisper -h
```

```
In [5]: import whisper # Whisper 라이브러리 불러오기
```

```
# 필요에 따라 모델 크기 설정 (예: "tiny", "small", "medium", "large")
model_name = "base" # 사용할 모델 이름 설정
try:
    # 모델 로드
    model = whisper.load_model(model_name)
except Exception as e:
    print(f"모델 '{model_name}' 로드 에러: {e}") # 오류 메시지 출력
    exit(1)

# 오디오 파일 경로 설정
audio_file_path = "WinstonChurch.ogg"
try:
    # 오디오 파일 필사(텍스트 변환)
    result = model.transcribe(audio_file_path)
    print('오디오의 텍스트 변환 결과:')
    print(result['text']) # 변환된 텍스트 출력
```

```
except Exception as e:  
    print(f"오디오 파일 '{audio_file_path}' 필사 에러: {e}") # 오류 메시지 출력
```

```
c:\ProgramData\anaconda3\Lib\site-packages\whisper\transcribe.py:132: UserWarning: FP16 is not supported on CPU; using FP32 instead
```

```
warnings.warn("FP16 is not supported on CPU; using FP32 instead")
```

오디오의 텍스트 변환 결과:

I speak to you for the first time as Prime Minister in a solemn hour for the life of our country, of our empire, of our allies, and above all of the cause of freedom. A tremendous battle is raging in France and Flanders. The Germans by a remarkable combination of air bombing and heavily armored tanks have broken through the French defenses north of the Maginot line, and strong columns of their armored vehicles are ravaging the open country, which for the first day or two was without defenders. They have penetrated deeply and spread alarm and confusion in their tracks. Behind them there are now appearing infantry in lorries, and behind them again, the large masses are moving forward. The regroupment of the French armies to make head against and also to strike at this intruding wedge and have been proceeding for several days, largely assisted by the magnificent efforts of the Royal Air Force. We must not allow ourselves to be intimidated by the presence of the armored vehicles in unexpected places behind our lives. If they are behind our front, the French are also at many points fighting actively behind their. Both sides are therefore in an extremely dangerous position. And if the French army and our own army are well handled, and I believe they will be, if the French retain that genius for recovery and counterattack, which they have so long been famous, and if the British army shows the dogged endurance and solid fighting power, which they have been so many examples in the past, then a sudden transformation of the scene might spring into being. It would be foolish, however, to disguise the gravity of the hour. It would be still more foolish to lose heart and courage, or to suppose that well-trained, well-equipped armies, numbering three or four millions of men, can be overcome in the space of a few weeks or even months by a scoop or raid of mechanized vehicles, however formidable. We may look with confidence to the stabilization of the front in France, and to the general engagement of the masses, which will enable the qualities of the French and British soldiers to be matched squarely against those of their adversaries. For myself, I have invincible confidence in the French army and its leaders. Only a very small part of that splendid army has yet been heavily engaged, and only a very small part of France has yet been invaded. There is good evidence to show that practically the whole of the specialized and mechanized forces of the enemy have been already thrown into the battle, and we know that very heavy losses have been inflicted upon them. No officer or man, no brigade or division, which grapples at close quarters with the enemy, wherever encountered, can fail to make a worthy contribution to the general result. The armies must cast away the idea of resisting attack behind concrete lines or natural obstacles, and must realize that the mass break can only be regained by furious and unrelenting assault, and this spirit must not only animate the high command but must inspire every fighting man. In the air, often at serious odds, often at odds, to the two who thought overwhelming, we have been clawing down three or four to one of our enemies, and the relative balance of the British and German air forces is now considerably more favorable to us than at the beginning of the battle. In cutting down the German bombers, we are fighting our own battle, as well as that of France. My confidence in our ability to fight it out to the finish with the German air force has been strengthened by the fierce encounters which have taken place and are taking place. At the same time, our heavy bombers are striking lightly at the taproot of German mechanized power, and have already inflicted serious damage upon the oil refineries on which the Nazi effort to dominate the world directly depends. We must expect that as soon as stability is reached on the western front, the bulk of that hideous apparatus of aggression, which dashed Holland into ruin and slavery in a few days, will return to penance. I am sure I speak for all when I say we are ready to face it. To endure it and to retaliate against it, to any extent that the unwritten laws of war permit. There will be many men and many women in this island who, when the ordeal comes upon them and come it will, will feel comfort and even a pride that they are sharing the perils of our leaders at the front. Soldiers, sailors, an Airman, God bless them and are drawing away for them a part at least of the onslaught they have to bear. He is not this the appointed time for all to make the utmost exertions in their power. If the battle is to be won, we must provide our men with ever increasing quantities of the weapons and ammunition they need. We must have and have quickly more aeroplanes, more tanks, more shells, more guns. There is imperative need for the vital munitions. They increase our strength against the powerfully armed enemy. They replace the wastage of the obstinate struggle and the knowledge that wastage will, speedily, be replaced, enables us to draw more readily

upon our reserves and throw them in now that everything counts no matter. Our task is not only to win the battle but to win the war. After this battle in France abates its force, they will come the battle for our islands. For all that Britain is and all that Britain means, that will be the struggle. In that supreme emergency, we shall not hesitate to take every step, even the most drastic to call forth from our people the last ounce and the last inch of effort I wish they are capable. The interests of property, the hours of labour are nothing compared to the struggle for life and honour, for right and freedom to which we have vowed ourselves. I have received from the chiefs of the French Republic and in particular from its indomitable Prime Minister, Monsieur Rayneau, the most sacred pledges that whatever happens they will fight to the end, be it bitter or be it glorious, nay, if we fight to the end it can only be glorious. Having received its Majesty's Commission, I have formed an administration of men and women of every party and have almost every point of view. We have deferred and quarreled in the past that now one bond unites at all. The wage war until victory is won and never to surrender ourselves to servitude and shame, whatever the cost and the agony may be. This is one of the most striking periods in the long history of France and Britain. It is also beyond out the most sublime, side by side, unadded except by their kitten kin in the great dominions and by the wide empires which rest beneath their shields. Side by side the British and French peoples have advanced to rescue not only Europe but mankind from the foulest and most soiled, destroying tyranny which had ever darkened and stained the pages of history. Behind us, behind the armies and fleets of Britain and France gather a group of shattered states and bludgeoned races, the checks, the poles, the Norwegians, the Danes, the Dutch, the Belgians, upon all of whom the long night of barbarism will descend, unbroken even by a star of hope, unless we conquer as conquer we must, as conquer we shall. Today is Trinity Sunday. Century to go words were written to be a call and a spur to the faithful servants of truth and justice. Arm yourselves and be men of valor and be in readiness for the conflict, for it is better for us to perish in battle than to look upon the outrage of our nation and our altars as the will of God is in heaven even so let him do.

```
In [6]: # whisper_test.py
import whisper # Whisper 라이브러리 불러오기

# 필요에 따라 모델 설정 (예: "tiny", "small", "medium", "large")
model_name = "small" # 사용할 모델을 medium으로 설정

try:
    # 모델 로드
    model = whisper.load_model(model_name)
except Exception as e:
    print(f"모델 '{model_name}' 로드 에러: {e}") # 모델 로드 시 오류가 발생하면 에러 메시지 출력
    exit(1)

# 오디오 파일 경로 설정
audio_file_path = "WinstonChurch.ogg"

try:
    # 오디오 파일을 영어로 필사
    result = model.transcribe(
        audio_file_path,
```



```

        language="en" # 영어로 필사 설정
    )
    print("영어 변환 결과:")
    print(result["text"]) # 영어 필사 결과 출력

    # 오디오 파일을 중국어로 필사
    result = model.transcribe(
        audio_file_path,
        language="zh" # 중국어로 필사 설정
    )
    print("중국어 변환 결과:")
    print(result["text"]) # 중국어 필사 결과 출력

    # 오디오 파일을 일본어로 필사
    result = model.transcribe(
        audio_file_path,
        language="ja" # 일본어로 필사 설정
    )
    print("일본어 변환 결과:")
    print(result["text"]) # 일본어 필사 결과 출력

    # .. 기타 언어 필사 예제 추가 가능
except Exception as e:
    print(f"오디오 파일 '{audio_file_path}' 필사 에러: {e}") # 필사 중 오류 발생 시 에러 메시지 출력

```

```

100%|████████████████████████████████████████| 461M/461M [00:05<00:00, 85.9MiB/s]
c:\ProgramData\anaconda3\Lib\site-packages\whisper\transcribe.py:132: UserWarning: FP16 is not supported on CPU; using FP32 instead
warnings.warn("FP16 is not supported on CPU; using FP32 instead")

```

영어 변환 결과:

I speak to you for the first time as Prime Minister in a solemn hour for the life of our country, of our empire, of our allies, and above all of the cause of freedom. A tremendous battle is raging in France and Flanders. The Germans, by a remarkable combination of air bombing and heavily armored tanks, have broken through the French defenses north of the Maginot Line, and strong columns of their armored vehicles are ravaging the open country, which for the first day or two was without defenders. They have penetrated deeply and spread alarm and confusion in their tracks. Behind them there are now appearing infantry in lorries, and behind them again the large masses are moving forward. The regroupment of the French armies to make head against and also to strike at this intruding wedge has been proceeding for several days, largely assisted by the magnificent efforts of the Royal Air Force. We must not allow ourselves to be intimidated by the presence of these armored vehicles in unexpected places behind our lines. If they are behind our front, the French are also at many points fighting actively behind theirs. Both sides are therefore in an extremely dangerous position, and if the French army and our own army are well handled, as I believe they will be, if the French retain that genius for recovery and counterattack, which they have so long been famous, and if the British army shows the dogged endurance and solid fighting power, of which there have been so many examples in the past, then a sudden transformation of the scene might spring into being. It would be foolish, however, to disguise the gravity of the hour. It would be still more foolish to lose heart and courage, or to suppose that well-trained, well-equipped armies numbering three or four millions of men can be overcome in the space of a few weeks or even months by a scoop or raid of mechanized vehicles, however formidable. We may look with confidence to the stabilization of the front in France and to the general engagement of the masses, which will enable the qualities of the French and British soldiers to be matched squarely against those of their adversaries. For myself, I have invincible confidence in the French army and its leaders. Only a very small part of that splendid army has yet been heavily engaged, and only a very small part of France has yet been invaded. There is good evidence to show that practically the whole of the specialized and mechanized forces of the enemy have been already thrown into the battle, and we know that very heavy losses have been inflicted upon them. No office nor our men, no brigade or division, which grapples at close quarters with the enemy wherever encountered, can fail to make a worthy contribution to the general result. The armies must cast away the idea of resisting attack behind concrete lines or natural obstacles, and must realize that mastery can only be regained by furious and unrelenting assault, and this spirit must not only animate the high command, but must inspire every fighting man. In the air, often at serious odds, often at odds, give a true thought overwhelming, we have been clawing down three or four to one of our enemy, and the relative balance of the British and German air forces is now considerably more favorable to us than at the beginning of the battle. In cutting down the German bombers, we are fighting our own battle as well as that of France. My confidence in our ability to fight it out to the finish with the German air force has been strengthened by the fierce encounters which have taken place and are taking place. At the same time, our heavy bombers are striking nightly at the taproot of German mechanized power and have already inflicted serious damage upon the oil refineries on which the nasty effort to dominate the world directly depends. We must expect that as soon as stability is reached on the western front, the bulk of that hideous apparatus of aggression which dashed Holland into ruin and slavery in a few days will be turned upon us. I am sure I speak for all when I say we are ready to face it, to endure it and to retaliate against it to any extent that the unwritten laws of war permit. There will be many men and many women in this island who when the ordeal comes upon them and come it will, will feel comfort and even a pride that they are sharing the perils of our lads at the front, soldiers, sailors and airmen, God bless them, and are drawing away for them a part at least of the onslaught they have to bear. Is not this the appointed time for all to make the utmost exertions in their power? If the battle is to be won, we must provide our men with ever increasing quantities of the weapons and ammunition they need. We must have and have quickly more aeroplanes, more tanks, more shells, more guns. There is imperious need for these vital munitions. They increase our strength against the powerfully armed enemy. They replace the wastage of the obstinate struggle and the knowledge that wastage will speedily be replaced enables us to draw more readily upon our reserves a

nd throw them in now that everything counts so much. Our task is not only to win the battle but to win the war. After this battle in France abates its force, there will come the battle for our island. For all that Britain is and all that Britain means, that will be the struggle. In that supreme emergency, we shall not hesitate to take every step, even the most drastic, to call forth from our people the last ounce and the last inch of effort of which they are capable. The interests of property, the hours of labor, are nothing compared to the struggle for life and honor, for right and freedom to which we have bowed ourselves. I have received from the chiefs of the French Republic and in particular from its indomitable Prime Minister, Monsieur Reynot, the most sacred pledges that whatever happens, they will fight to the end, be it bitter or be it glorious. Nay, if we fight to the end, it can only be glorious. Having received his Majesty's commission, I have formed an administration of men and women of every party and of almost every point of view. We have differed and quarreled in the past that now one bond unites at all to wage war until victory is won and never to surrender ourselves to servitude and shame, whatever the cost and the agony may be. This is one of the most awe-striking periods in the long history of France and Britain. It is also beyond doubt the most sublime, side by side, unedited except by their kith and kin in the great dominions and by the wide empires which rest beneath their shield. Side by side, the British and French peoples have advanced to rescue not only Europe but mankind from the foulest and most sole destroying tyranny which has ever darkened and stained the pages of history. Behind them, behind us, behind the armies and feats of Britain and France gather a group of shattered states and bludgeoned races, the Czechs, the Poles, the Norwegians, the Danes, the Dutch, the Belgians, upon all of whom the long night of barbarism will descend, unbroken even by a star of hope, unless we conquer as conquer we must, as conquer we shall. Today is Trinity Sunday. Century to go words were written to be a call and a spur to the faithful servants of truth and justice. Arm yourselves and be men of valor and be in readiness for the conflict, for it is better for us to perish in battle than to look upon the outrage of our nation and our altars, as the will of God is in heaven. Even so, let him do.

중국어 변환 결과:

今天我嚴格地槽OLM為 日盤黨國民當選營幕為香港人為一條一位重愛了全法玳沒通修回去你有別的純粹以西風土壤之他們的大不一樣的軍事也品刊的單元無論他們採訪日報 打工兵的軍隊還有他們的不斷 archive責佈英安們的形式改善攻陷不駕快地這場渣雅和儀式四漫則面度高我們對可以相不支持良心的出來世界下 조그很隨意本海樹物 Devil Gün.Jway,雷尾克至眼 所有劍厚 連 我 更 誰 量 樂 wLB他們都有當過永遠 running 人手要足以為解人手上面也不會因為還會更大的笨拙在脫佔押肺炎這是量一點的用手衝勁玄關獨特期間卡到浪 coriander而向大自然的打擊方法將會扔飛輸出壞的能力和富裕人所有的武器都幫助右 British soldiers去硬度對這些子女經驗我藉ي的任命他們可以得到 judge game剪掉在近來已經要大大地是 fanvee優點正常而家晃然的訴求 Survey而且想 PB太陽但不知道美國他認為音樂修 come助量快速要台我們已被量責取從3,4,1%而ены因此民體網民收斂 fätt作攻擊比之最接近我們的火力現在都更滿意在要阻止高 YouTube網民發訊息我們採訪我們的獨有女戰也與會歷史奧斯利庫利我們堅持英國的資深讓英國公報和攻擊才ups-ed 而如此微粗的否認到這些事ality卻同時 五重的打擊此ジ盟的企業民心他們是有意對高壓力的十 concerning高攻擊旗艦噓正虛伽此各地的 sister第三他的職業垃圾演員恰在被刷淨的暴ア我們 decision at Tool who是項目19警方呵呵呵 dear很棒好很棒很棒 c offins你好 end quarter如果戰鬥要成功我們必須給予我們人們每次增加的量量武器和武器他們需要我們必須要得到快速的更多空軍 更多艙更多槍更多槍他們必須要有更多的武器他們必須要增加我們的力量對我們強烈的對戰他們必須要兌換兌換的兌換的兌換而且我們必須要把能力兌換的兌換的兌換能夠更加快速地兌換我們的兌換的兌換我們必須要兌換的兌換現在我們能夠兌換的兌換我們的目標不是要贏戰鬥但要贏戰鬥在這戰鬥中在法國的戰鬥中我們會來戰鬥的在全球的是全球的我們必須要兌換的兌換在這次的我們必須要每一步都要向我們人們打招呼我們最終的和最終的能力他們必須要的他們的興趣他們的時間他們的對戰他們的命和 honour他們的自由我們必須要兌換的兌換的兌換我們必須要兌換的兌換他們的他們的我們他們的兌換他們的英國他們的英國他們的團隊他們的國籍他們的軍隊他們的國籍他們的日子他們的日子他們的德國他們的德國他們的所有人他們的一天的他們的一天的

일본어 변환 결과:

ini喜歡 diedモンがいるvood st2翻 aluminum ||2あるいは期待している周会の踏まえのない defiって回去に Ga Mainlyの所第一イネが人間ピンバリッハ発売紫分八連座へメガネの意識inov総の forward夏日良くなったこれで 希望は、お伝えしながら君にとってくるぜび cling に近くへ 務めや揺る私のは様々な側面だありがともに、 ads リア平専美彼女の攻撃を信じてally 立ち来了私は、私の代償は中華装備でその險ねえ同

レクス準の皆さんについてST 食事が始まるそして、ブリッシュアーメイドが強くなった場合、その場合、その前に、その場合、突然の真似を見ると、それが、それが、それが、それが、それが、それが、それが、それが、郵endの他部の flying 列工船 deck 愛してエリア金のも聯の but ぎやつゆれどものシーバーセット仲間戦術の浮かぶ地位の earth 飯 さんでも縦ス造る配置への なによってまそいぜるそれに応著して go 本載したうる She なん それとしてはがふうなし てか脱出 twisted 分かったシナシ freshlyでいます〜彼はきれいに調整されたものでそれ bienPinci Army steal するのほうれしいもですな日カードがippen ペシャ painful anythingどんな中の男性がそれに了のよう💎💎 lays upon them ノー車の建ての子 Ling文つけ者ックア升谷どこから再💎仮別行う コントロウ味喰notと告诉之下だ分似合わせの触っています washing of 丁寧hoミッ清雄 pick カーニングできるや take power 以上映像明るかったアーフ light o波打消口が中霊してるのが発電しました必要でアルバガイスはOpen すちは白が发展した 内も孫秋 roller 下それはグリーバーマースは グレーバーマースの力を攻撃することを拡大することですそして 強力の攻撃を 攻めたことがなぜ 世界を 直径に 勝手に 受け入れられることを期待していますこれを ウェステンプラントの 速さに 発展し 信じられた スターボリューションを

위스퍼 텍스트 변환 최적화 기법

목차

- 오픈AI 음성-텍스트 API 사용하기
- 위스퍼 텍스트 변환 개선하기

```
In [8]: # !pip install httpx
```

```
In [ ]: from openai import OpenAI
import os
from dotenv import load_dotenv
load_dotenv()
load_dotenv(override=True)

OPENAI_API_KEY = os.getenv("OPENAI_API_KEY")
```

```
In [10]: client = OpenAI()

# 오디오 파일 경로 정의
audio_file_path = os.path.abspath("Winston_Church.ogg")

# 사용할 모델 정의
model = "whisper-1" # Whisper-1 모델 사용

# 오디오 파일을 바이너리 모드로 열기
```

```
with open(audio_file_path, 'rb') as audio_file:
    # 오디오 파일 전사(텍스트 변환)
    transcript = client.audio.transcriptions.create(
        model=model,
        file=audio_file
    )
    print(transcript.text) # 전사된 텍스트 출력
```

I speak to you for the first time as Prime Minister in a solemn hour for the life of our country, of our empire, of our allies, and above all of the cause of freedom. A tremendous battle is raging in France and Flanders. The Germans, by a remarkable combination of air bombing and heavily armoured tanks, have broken through the French defences north of the Maginot Line, and strong columns of their armoured vehicles are ravaging the open country, which for the first day or two was without defenders. They have penetrated deeply and spread alarm and confusion in their track. Behind them there are now appearing infantry in lorries, and behind them again the large masses are moving forward. The regroupment of the French armies to make head against, and also to strike at, this intruding wedge has been proceeding for several days, largely assisted by the magnificent efforts of the Royal Air Force. We must not allow ourselves to be intimidated by the presence of these armoured vehicles in unexpected places behind our lines. If they are behind our front, the French are also at many points fighting actively behind theirs. Both sides are therefore in an extremely dangerous position, and if the French army and our own army are well handled, as I believe they will be, if the French retain that genius for recovery and counter-attack for which they have so long been famous, and if the British army shows the dogged endurance and solid fighting power of which there have been so many examples in the past, then a sudden transformation of the scene might spring into being. It would be foolish, however, to disguise the gravity of the hour. It would be still more foolish to lose heart and courage, or to suppose that well-trained, well-equipped armies, numbering three or four millions of men, can be overcome in the space of a few weeks, or even months, by a scoop or raid of mechanised vehicles, however formidable. We may look with confidence to the stabilisation of the front in France, and to the general engagement of the masses, which will enable the qualities of the French and British soldiers to be matched squarely against those of their adversaries. For myself, I have invincible confidence in the French army and its leaders. Only a very small part of that splendid army has as yet been heavily engaged. And only a very small part of France has yet been invaded. There is good evidence to show that practically the whole of the specialised and mechanised forces of the enemy have been already thrown into the battle, and we know that very heavy losses have been inflicted upon them. No officer or man, no brigade or division, which grapples at close quarters with the enemy wherever encountered, can fail to make a worthy contribution to the general result. The armies must cast away the idea of resisting attack behind concrete lines or natural obstacles, and must realise that mastery can only be regained by furious and unrelenting assault. And this spirit must not only animate the high command, but must inspire every fighting man. In the air, often at serious odds, often at odds, hitherto thought overwhelming, we have been clawing down three or four to one of our enemies. And the relative balance of the British and German air forces is now considerably more favourable to us than at the beginning of the battle. In cutting down the German bombers, we are fighting our own battle, as well as that of France. My confidence in our ability to fight it out to the finish with the German air force has been strengthened by the fierce encounters which have taken place and are taking place. At the same time, our heavy bombers are striking nightly at the taproot of German mechanised power, and have already inflicted serious damage upon the oil refineries on which the Nazi effort to dominate the world directly depends. We must expect that as soon as stability is reached on the western front, the bulk of that hideous apparatus of aggression which dashed Holland into ruin and slavery in a few days will be turned upon us. I am sure I speak for all when I say we are ready to face it, to endure it, and to retaliate against it to any extent that the unwritten laws of war permit. There will be many men and many women in this island who, when the ordeal comes upon them, and it will, will feel comfort and even a pride that they are sharing the perils of our lads at the front, soldiers, sailors, and airmen, God bless them, and are drawing away for them a part at least of the onslaught they have to bear. Is not this the appointed time for all to make the utmost exertions in their power? If the battle is to be won, we must provide our men with ever increasing quantities of the weapons and ammunition they need. We must have, and have quickly, more aeroplanes, more tanks, more shells, more guns. There is imperative need for these vital munitions. They increase our strength against the powerfully armed enemy, they replace the wastage of the obstinate struggle, and the knowledge that wastage will speedily be replaced enables us to draw more readily upon our reserves and throw them in now that everything counts so much. Our task is not only to win the battle, but to win the war. After

this battle in France abates its force, there will come the battle for our islands, for all that Britain is, and all that Britain means. That will be the struggle. In that supreme emergency, we shall not hesitate to take every step, even the most drastic, to call forth from our people the last ounce and the last inch of effort of which they are capable. The interests of property, the hours of labor, are nothing compared to the struggle for life and honor, for right and freedom to which we have bowed ourselves. I have received from the chiefs of the French Republic, and in particular from its indomitable Prime Minister, M. Reynaud, the most sacred pledges that whatever happens, they will fight to the end, be it bitter or be it glorious. Nay, if we fight to the end, it can only be glorious. Having received His Majesty's commission, I have formed an administration of men and women of every party, and of almost every point of view. We have differed and quarreled in the past, but now one bond unites us all, to wage war until victory is won, and never to surrender ourselves to servitude and shame, whatever the cost and the agony may be. This is one of the most awe-striking periods in the long history of France and Britain. It is also, beyond doubt, the most sublime. Side by side, unaided except by their kith and kin in the great dominions, and by the wide empires which rest beneath their shield, side by side, the British and French peoples have advanced to rescue not only Europe, but mankind, from the foulest and most soul-destroying tyranny which has ever darkened and stained the pages of history. Behind them, behind us, behind the armies and fleets of Britain and France, gather a group of shattered states and bludgeoned races—the Czechs, the Poles, the Norwegians, the Danes, the Dutch, the Belgians—upon all of whom the long night of barbarism will descend, unbroken even by a star of hope. Unless we conquer, as conquer we must, as conquer we shall. Today is Trinity Sunday. Centuries ago, words were written to be a call and a spur to the faithful servants of truth and justice. Arm yourselves and be ye men of valor, and be in readiness for the conflict, for it is better for us to perish in battle than to look upon the outrage of our nation and our altars, as the will of God is in heaven. Even so, let him do.

```
In [11]: # 오디오 파일 경로 정의
audio_file_path = os.path.abspath("Winston_Church.ogg")

# 사용할 모델 정의
model = "whisper-1" # Whisper-1 모델 사용

# 오디오 파일을 바이너리 모드로 열기
with open(audio_file_path, 'rb') as audio_file:
    # 오디오 파일 번역 및 텍스트 변환
    transcript = client.audio.translations.create(
        model=model,
        file=audio_file,
    )
    print(transcript.text) # 번역된 텍스트 출력
```

I speak to you for the first time as Prime Minister in a solemn hour for the life of our country, of our empire, of our allies, and above all of the cause of freedom. A tremendous battle is raging in France and Flanders. The Germans, by a remarkable combination of air bombing and heavily armoured tanks, have broken through the French defences north of the Maginot Line, and strong columns of their armoured vehicles are ravaging the open country, which for the first day or two was without defenders. They have penetrated deeply and spread alarm and confusion in their track. Behind them there are now appearing infantry in lorries, and behind them again the large masses are moving forward. The regroupment of the French armies to make head against, and also to strike at, this intruding wedge has been proceeding for several days, largely assisted by the magnificent efforts of the Royal Air Force. We must not allow ourselves to be intimidated by the presence of these armoured vehicles in unexpected places behind our lines. If they are behind our front, the French are also, at many points, fighting actively behind theirs. Both sides are therefore in an extremely dangerous position, and if the French army and our own army are well handled, as I believe they will be, if the French retain that genius for recovery and counter-attack for which they have so long been famous, and if the British army shows the dogged endurance and solid fighting power of which there have been so many examples in the past, then a sudden transformation of the scene might spring into being. It would be foolish, however, to disguise the gravity of the hour. It would be still more foolish to lose heart and courage, or to suppose that well-trained, well-equipped armies, numbering three or four millions of men, can be overcome in the space of a few weeks or even months by a scoop or raid of mechanised vehicles, however formidable. We may look with confidence to the stabilisation of the front in France, and to the general engagement of the masses, which will enable the qualities of the French and British soldiers to be matched squarely against those of their adversaries. For myself, I have invincible confidence in the French army and its leaders. Only a very small part of that splendid army has yet been heavily engaged, and only a very small part of France has yet been invaded. There is good evidence to show that practically the whole of the specialised and mechanised forces of the enemy have been already thrown into the battle, and we know that very heavy losses have been inflicted upon them. 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There will be many men and many women in this island who, when the ordeal comes upon them, and come it will, will feel comfort and even a pride that they are sharing the perils of our lads at the front, soldiers, sailors, and airmen, God bless them, and are drawing away from them a part at least of the onslaught they have to bear. Is not this the appointed time for all to make the utmost exertions in their power? If the battle is to be won, we must provide our men with ever-increasing quantities of the weapons and ammunition they need. We must have, and have quickly, more airplanes, more tanks, more shells, more guns. There is imperious need for these vital munitions. They increase our strength against the powerfully armed enemy, they replace the wastage of the obstinate struggle, and the knowledge that wastage will speedily be replaced enables us to draw more readily upon our reserves and throw them in now that everything counts so much. 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r this battle in France abates its force, there will come the battle for our islands, for all that Britain is, and all that Britain means. That will be the struggle. In that supreme emergency, we shall not hesitate to take every step, even the most drastic, to call forth from our people the last ounce and the last inch of effort of which they are capable. The interests of property, the hours of labor, are nothing compared to the struggle for life and honor, for right and freedom to which we have vowed ourselves. I have received from the Chiefs of the French Republic, and in particular from its indomitable Prime Minister, Monsieur Reynaud, the most sacred pledges that whatever happens, they will fight to the end, be it bitter or be it glorious. Nay, if we fight to the end, it can only be glorious. Having received His Majesty's commission, I have formed an administration of men and women of every party and of almost every point of view. We have differed and quarreled in the past, but now one bond unites us all, to wage war until victory is won, and never to surrender ourselves to servitude and shame, whatever the cost and the agony may be. This is one of the most awe-striking periods in the long history of France and Britain. It is also, beyond doubt, the most sublime. Side by side, unaided except by their kith and kin in the great dominions, and by the wide empires which rest beneath their shield, side by side the British and French peoples have advanced to rescue not only Europe but mankind from the foulest and most soul-destroying tyranny which has ever darkened and stained the pages of history. Behind them, behind us, behind the armies and fleets of Britain and France, gather a group of shattered states and bludgeoned races, the Czechs, the Poles, the Norwegians, the Danes, the Dutch, the Belgians, upon all of whom the long night of barbarism will descend, unbroken even by a star of hope, unless we conquer as conquer we must, as conquer we shall. Today is Trinity Sunday. Centuries ago, words were written to be a call and a spur to the faithful servants of truth and justice. Arm yourselves and be ye men of valour, and be in readiness for the conflict, for it is better for us to perish in battle than to look upon the outrage of our nation and our altars, as the will of God is in heaven, even so let him do.

위스퍼 텍스트 변환 개선하기

데이터 준비

```
!curl -o cypher.mp3 https://raw.githubusercontent.com/LDJWJ/OpenAIGPTForPythonDevelopersFiles/main/audio/cypher.mp3
```

```
In [12]: # 오디오 파일 다운로드
!curl -o cypher.mp3 https://raw.githubusercontent.com/LDJWJ/OpenAIGPTForPythonDevelopersFiles/main/audio/cypher.mp3
```

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload Upload	Total	Spent	Left	Speed
0	0	0	0	0	0	0	0
100	81041	100	81041	0	0	285k	0
							287k

```
In [13]: # 오디오 파일 경로 정의
audio_file_path = os.path.abspath(
    "cypher.mp3" # 오디오 파일 경로를 설정합니다.
```

```

)

# 모델 정의
model = "whisper-1" # Whisper-1 모델을 사용합니다.

# 프롬프트 정의
prompt = "Cypher" # 올바른 철자를 힌트로 제공하기 위해 'Cypher'를 프롬프트로 설정합니다.

# 오디오 파일을 바이너리 모드로 열기
with open(audio_file_path, 'rb') as audio_file:
    # 오디오 파일 전사
    transcript = client.audio.transcriptions.create(
        model=model,
        file=audio_file,
        prompt=prompt
    )

# 전사된 텍스트 출력
print(transcript.text)

```

We need an exit. You're not far from Cypher. Cypher?

```

In [ ]: # 오디오 파일 경로 정의
audio_file_path = os.path.abspath(
    "cypher.mp3" # 오디오 파일의 경로를 설정합니다.
)

# 모델 정의
model = "whisper-1" # Whisper-1 모델을 사용합니다.

# 오디오 파일을 바이너리 모드로 열기
with open(audio_file_path, 'rb') as audio_file:
    # 오디오 파일 전사(Transcription)
    transcript = client.audio.transcriptions.create(
        model=model,
        file=audio_file
    )

# GPT-4.1-mini를 사용하여 텍스트 후처리(post-processing)
system_prompt = """당신은 영화 전문가입니다.
오디오 파일의 전사를 수정하는 것이 당신의 역할입니다.

```

예를 들어, 'Cipher' 또는 'Cypher'의 단어는 'Cyber'로 수정해야 합니다."""

모델 정의

model = "gpt-4.1-mini" # GPT-4o 모델을 사용합니다.

텍스트 후처리

```
response = client.chat.completions.create(
    model=model,
    messages=[
        {
            "role": "system",
            "content": system_prompt # 시스템 프롬프트로 후처리 작업을 설정합니다.
        },
        {
            "role": "user",
            "content": transcript.text # Whisper 전사 텍스트를 사용자 입력으로 전달합니다.
        }
    ]
)
```

수정된 텍스트 출력

print(response.choices[0].message.content) # GPT-4.1-mini로 수정한 텍스트를 출력합니다.

We need an exit. You're not far from Cyber. Cyber?